

Joongwon Chae

Training-Free Anomaly Detection · Memory-Augmented Inference · Medical AI

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RESEARCH INTERESTS

I develop training-free and gradient-free inference methods that adapt frozen foundation-model representations through memory, retrieval, prompting, routing, and calibration — avoiding the retraining, decoders, dense labels, and human prompts these models usually require. My work centers on anomaly detection and medical image analysis under limited labels, retraining budgets, and human interaction.

EDUCATION

Tsinghua University — Shenzhen International Graduate School (SIGS) 2024 – 2027 (expected)
M.S., Institute of Biopharmaceutical & Health Engineering (IBHE) · Advisor: Prof. Peiwu Qin · Shenzhen, China

Shanghai Jiao Tong University (SJTU) — School of Automation 2018 – 2024
B.S. in Automation · Shanghai, China

RESEARCH EXPERIENCE

Graduate Researcher — Tsinghua University (SIGS) 2024 – present
Independently leading method development under the supervision of Prof. Peiwu Qin.

- **Training-free anomaly detection (ProCon, GCR, StructCore).** Led three first-author works end-to-end — formulating the research questions, designing and implementing the methods, running the experiments, and drafting the manuscripts: soft projection-consistency memory (ProCon), geometry-consistent cross-head routing for continual AD (GCR), and structure-aware map-to-score calibration (StructCore). ProCon reaches 99.8%, 99.2%, and 93.2% image-level AUROC on MVTec-AD, VisA, and Real-IAD with a frozen backbone and no training.
- **Human-prompt-free medical segmentation (Memory-SAM).** Designed a retrieval-to-prompt method turning frozen DINOv3 correspondence into automatic SAM2 prompts; 0.984 and 0.973 mIoU on the public HIT-Tongue dataset and a 2,155-image smartphone benchmark.
- Owned the full research cycle end-to-end — problem formulation, method design, implementation, experiments and ablations, and manuscript writing — with open-source releases (github.com/jw-chaе).

Clinical ML study — gallbladder-cancer T-staging 2024 – 2025

- First-authored a study (published in *Frontiers in Oncology*) showing a reasoning LLM reaches 89.6% accuracy for pre-operative T-staging from radiology text, vs. near-chance blood-biomarker ML, across 232 patients.
- Designed the full evaluation — classical ML baselines, SMOTE analysis, AUROC with 1,000-sample bootstrap confidence intervals — and led the writing.

Multimodal medical-AI systems (TCM) — co-first author 2024 – present

- Co-led ViTCM-LLM (IEEE BIBM 2025) and MMIR-TCM: co-developed the MedTCM multimodal dataset, designed the TDEU evaluation metric, and contributed the retrieval-augmented reasoning and safety-constrained pipeline. The domain-adapted system outperformed zero-shot GPT-4o and Gemini 2.5 Flash on the MedTCM evaluation protocol.

PEER-REVIEWED PUBLICATIONS (* equal contribution; bold = me)

1. **J. Chae**, Z. Wang, D. Wu, L. Zhang, A. Tuzikov, M. T. Madiyevich, M. Xu, D. Yu, P. Qin. "Pre-operative T-stage discrimination in gallbladder cancer using machine learning and DeepSeek-R1." *Frontiers in Oncology*, 2025.
doi.org/10.3389/fonc.2025.1613462
2. L. Luo*, **J. Chae***, Y. Liu, I. Pantic, V. Devedzic, Z. Sun, Z. Zeng, S. Matlatipov, X. Yin, P. Qin. "ViTCM-LLM: A Multimodal RAG Framework for Advanced TCM Clinical Decision Support." *IEEE Int. Conf. on Bioinformatics and Biomedicine (BIBM)*, 2025. **Co-first**
[DOI](#) [Code](#)

PREPRINTS & MANUSCRIPTS UNDER REVIEW

3. **J. Chae**, et al. "ProCon: Projection-Consistency Memory for Training-Free Anomaly Detection." *Under review at AAAI 2027*.
arXiv:2607.04894 Code
4. **J. Chae**, L. Luo, Y. Liu, et al. "GCR: Geometry-Consistent Routing for Task-Agnostic Continual Anomaly Detection." *Under review at AAAI 2027*.
arXiv:2601.01856 Code
5. **J. Chae**, et al. "StructCore: Structure-Aware Image-Level Scoring for Training-Free Unsupervised Anomaly Detection." *Under review at WACV 2027*.
arXiv:2602.17048 Code
6. **J. Chae**, L. Luo, X. Yuan, et al. "Memory-SAM: Human-Prompt-Free Tongue Segmentation via Retrieval-to-Prompt." *Under review at AAAI 2027*.
arXiv:2510.15849 Code
7. L. Luo*, **J. Chae***, et al. "MMIR-TCM: Memory-Integrated Multimodal Inference and Retrieval for TCM Clinical Decision Support." *Under review at IEEE J. Biomedical & Health Informatics (JBHI)*. **Co-first**
arXiv:2607.01814
8. **J. Chae**, R. Wang, et al. "Beyond Curve Fitting: Neuro-Symbolic Agents for Context-Aware Epidemic Forecasting." *Under review at IEEE BIBM 2026*.
arXiv:2511.23276
9. L. Luo*, **J. Chae***, et al. "ViGen: Video-based Generation with GRPO for Dynamic Image Editing." *Under review at AAAI 2027*. **Co-first**

HONORS & SCHOLARSHIPS

- **Excellent Student Scholarship, First Class** (International Students), Tsinghua University, 2024
- **Yingcai First-Class Scholarship**, Institute of Biopharmaceutical & Health Engineering, Tsinghua University, 2025 — *awarded to the top-ranked international student in the department*
- **Tsinghua University International Graduate Tuition Scholarship**, 2025–2026
- **Shenzhen Universiade International Scholarship Foundation (SUISF) Scholarship**, 2026

INDUSTRY RESEARCH EXPERIENCE

Ratel Soft — Research Intern Jul 2025 – present

- Developing vision-based anomaly-detection pipelines for collaborative-robot (cobot) assembly-line inspection.
- Benchmarking frozen foundation-model features against task-specific baselines; building data and evaluation tooling in PyTorch.

TECHNICAL SKILLS

- **Programming:** Python, PyTorch, scikit-learn, OpenCV
- **Foundation models:** DINOv2/v3, SAM/SAM2, CLIP; Hugging Face Transformers, PEFT/LoRA, LLaMA-Factory
- **Systems:** Docker, DeepSpeed, multi-GPU training, RAG pipelines, industrial machine-vision cameras

LANGUAGES

Korean (native) · English (professional working proficiency) · Chinese (HSK 6) · Japanese (JLPT N2)